DEEP LEARNING LAB

Algorithms for Selected Experiments

Experiment 1 — Regression

Algorithm

Step 1: Start.
Step 2: Import the required libraries and load the Advertising dataset.
Step 3: Select TV advertising as the input feature and Sales as the target for Simple Linear Regression.
Step 4: Split the data into training and testing sets (80/20, random_state = 42).
Step 5: Train a Linear Regression model, predict Sales, and calculate MSE and R^2 score. Plot the data and regression line.
Step 6: Select TV, Radio, and Newspaper as input features for Multiple Linear Regression.
Step 7: Train the model, predict Sales, calculate MSE and R^2 score, and plot actual versus predicted Sales.
Step 8: Transform the TV feature into polynomial features of degree 2.
Step 9: Train the Polynomial Regression model, predict Sales, calculate MSE and R^2 score, and plot the polynomial curve.
Step 10: Display the results and stop.

Experiment 2 — K-Nearest Neighbors (KNN) Classification

Algorithm

Step 1: Start.
Step 2: Import the required libraries and load the Iris dataset.
Step 3: Standardize the features using StandardScaler.
Step 4: Split the dataset into training and testing sets (80/20, random_state = 42).
Step 5: Create a KNN classifier with K = 5.
Step 6: Train the classifier using the training data.
Step 7: Predict the class labels of the test data.
Step 8: Calculate and display accuracy, macro precision, and macro recall.
Step 9: Stop.

Experiment 3 — Naive Bayes Classification (NBC)

Algorithm

Step 1: Start.
Step 2: Import the required libraries and load the Iris dataset.
Step 3: Separate the features and target labels.
Step 4: Standardize the features using StandardScaler.
Step 5: Split the data into training and testing sets (80/20, random_state = 42).
Step 6: Create and train a Gaussian Naive Bayes classifier.
Step 7: Predict the classes of the test samples.
Step 8: Calculate and display the accuracy.
Step 9: Stop.

Experiment 5 — Hierarchical Agglomerative Clustering (HAC)

Algorithm

Step 1: Start.
Step 2: Import the required libraries, generate 150 sample points with three cluster centers, and standardize the data.
Step 3: Compute the hierarchical linkage matrix using Ward linkage.
Step 4: Plot the dendrogram to show the order in which clusters are merged.
Step 5: Create an Agglomerative Clustering model with three clusters and Ward linkage.
Step 6: Fit the model and obtain a cluster label for each data point.
Step 7: Plot the data points using different colors for the assigned clusters.
Step 8: Display the dendrogram and clustering plot, then stop.

Experiment 6 — K-Means Clustering

Algorithm

Step 1: Start.
Step 2: Import the required libraries, generate 150 sample points with three cluster centers, and standardize the data.
Step 3: Create a K-Means model with K = 3 and fit it to assign each data point to a cluster.
Step 4: Run K-Means for K = 1 to 9, record the inertia (within-cluster sum of squares) of each, and plot the Elbow curve.
Step 5: Obtain the cluster centroids of the K = 3 model.
Step 6: Plot the clusters in different colors and mark the centroids.
Step 7: Display the plot and stop.